0203

ALGEBRA I PRACTICE TEST

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0204

Part II

Answer all 6 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit.

25 Solve the equation below algebraically.

52/15 16/7

$$6 - \frac{2}{3}(x + 5) = 4x$$

$6 - \frac{2}{3}x - 3\frac{1}{3} = 4x$

$6 - 3\frac{1}{3} = 4\frac{2}{3}x$

$\frac{2\frac{2}{3}}{4\frac{2}{3}} = \frac{4\frac{2}{3}x}{4\frac{2}{3}}$

$\boxed{x = \frac{4}{7}}$

answer

26 $F = 4x^2 - 3x + 1$ and $G = -2x^2 + 5$. Find the trinomial that represents H, if $H = 3G - F$.

$3G = 3(-2x^2 + 5)$

$3G = -6x^6 + 15$

$3G - F = (-6x^6 + 15) - (4x^2 - 3x + 1)$

$\downarrow$

$\boxed{10x^4 + 3x + 16}$

$\downarrow$

H

0205

27 Algebraically determine the zeros of the function $f(x) = x^2 - 4x + 3$.

Standard: $ax^2 - 4x + 3$

Factored: $(x - m)(x - n)$

	x	-3
x	x^2	$-3x$
-1	$-x$	3

$\rightarrow -4x$

$\downarrow$

$x^2 - 4x + 3$

$\downarrow$

$F(x) = (x-3)(x-1)$

$\downarrow$

$\boxed{x = 3}$

$\boxed{x = 1}$

> Both zeros

28 In February 2021, the state of Texas saw one of the worst snowstorms in its history. Snowfall totals recorded in Dallas, TX are shown in the table below.

1am to noon $= \frac{32}{11} = 2.9$

6am to 3pm $= \frac{25}{9} = 2.77$

Dallas, TX	
Time	**Snow** (inches)
1 a.m.	1
3 a.m.	5
6 a.m.	11
12 noon	33
3 p.m.	36

Which of the following intervals had the greatest rate of snowfall, in inches per hour? 1 a.m. to 12 noon or 6 a.m. to 3 p.m. Justify your answer.

The 1 a.m to 12 noon interval had the greatest rate of snowfall. From that interval the rate of snowfall was 2.9 inches. From 6 a.m to 3 p.m the rate of snowfall is 2.77 inches. 2.9 > 2.77, so it is a greater rate.

29 The gravitational force between two objects can be calculated using the formula $F_g = \frac{GM_1M_2}{r^2}$. In the formula, G is called the gravitational constant. The variables M_1 and M_2 represent the masses of the two objects between which the force is being measured. Lastly, r is the distance between the objects. Solve for the positive value of r, in terms of the other variables.

$$F_g = \frac{GM_1M_2}{r^2}$$

$$\downarrow$$

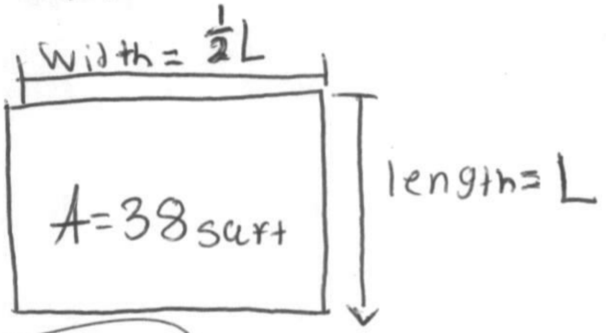
$$\boxed{r = \sqrt{\frac{GM_1M_2}{F_g}}}$$

- Square root to get rid of the exponent

0206

30 Four Seasons Landscaping is creating a rectangular flower bed at a customer's home. The width of the flower bed will be half of the length. The total area of the flower bed is 38 square feet. Write and solve an equation to determine the width of the flower bed to the *nearest tenth* of a foot.

$$\boxed{\frac{A}{L} = W}$$



$\boxed{w = 4.5 \text{ ft}}$

Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

0207

31 A couple planning a backyard wedding is researching different tent rental companies for their event. They contact two different companies. Buffalo Party Rental charges a $50 set-up fee, and $5 per square foot of tent space. Tent Genie charges a $25 set-up fee, and $6 per square foot of tent space.

Write an equation for Buffalo Party Rental that could be used to determine the total cost B, when x square feet of tent space is ordered. Write a second equation for Tent Genie that could be used to determine the total cost T, when x square feet of tent space is ordered.

Buffalo Party

f(X) = T

$\boxed{B = \$50 + \$5X}$

→ Buffalo Party

Tent Genie

• 25 → set up fee

• 6 → rate

$\boxed{T = \$25 + \$6x}$ → Tent genie

Determine algebraically and state the *minimum* square feet, rounded to the *nearest square foot*, of tent space that must be ordered for Buffalo Party Rental to be the cheaper option.

$$\frac{\$50 - \$25 = \$25}{6x - 5x = X}$$

would be equal/more after 25

↓

25 + 1 = 26 → minimum for Buffalo to be cheaper

B = $50 + (5 · 26)

B = 180

T = $25 + (6 · 26)

T = 181

181 > 180

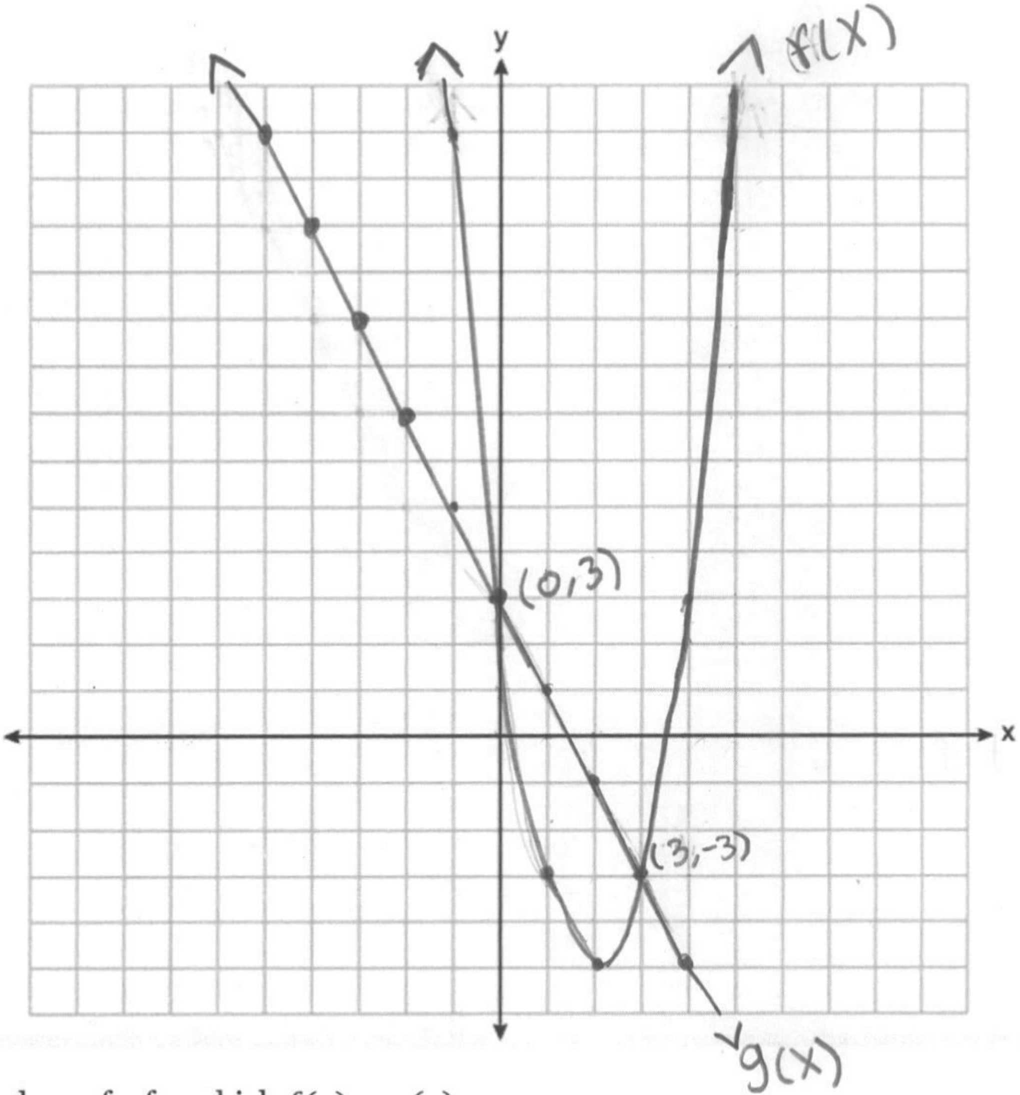
The minimum would be 26 square feet

32 Graph the two functions on the coordinate plane below.

y=mx+b

$f(x) = 2x^2 - 8x + 3$

$g(x) = -2x + \underline{3} \rightarrow$ y-int



Find all values of x for which $f(x) = g(x)$.

f(X)	g(X)
X=0, y=3	(0,3)
X=3, y=-3	(3,-3)

When X is 0 and 3, the functions f(X) and g(X) are equal.

33 The table below shows the number of hours ten different scholars at Marion P. Thomas Middle School spent studying and their scores on a recent history test.

Hours Spent Studying (x)	0	1	2	4	4	4	6	6	7	8
Test Scores (y)	35	40	46	65	67	70	82	88	82	95

Write the linear regression for this data set. Round all values to the *nearest hundredth.*

$\boxed{y = 7.79x + 34.27}$

State the correlation coefficient rounded to the *nearest hundredth.*

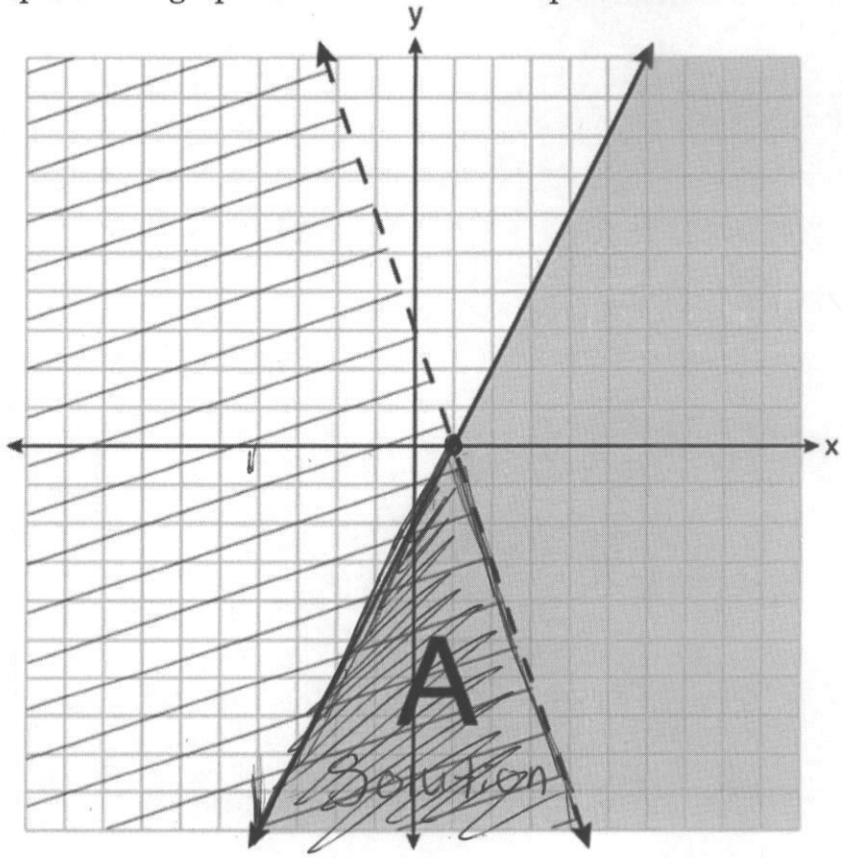
$r = .98$

The correlation coefficient is .98.

Explain what the correlation coefficient suggests in the context of the problem.

The correlation coefficient suggest that the data set is strong and close to being a straight line.

34 A system of inequalities is graphed on the coordinate plane below.



Write the system of inequalities that is represented by the graph above.

A = X to 1

9 to -4, 4 Range=8

What does region A represent?

Region A represents the Solution.

What does the gray region represent?

The gray represents the Solution of 1 Side of the functions.

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

0209

35 The CoinStar machine at the local supermarket is broken. Instead of providing the exact number of each type of coin that is deposited, the machine can only display the total number of coins and total value. The machine shows 90 coins, with a value of \$17.55.

If Mr. Rogers only placed dimes and quarters in the machine, write a system of equations in two variables, or an equation in one variable that could be used to model this situation.

total = \$17.55

d = dime = \$.10

q = quarter = \$.25

$\boxed{17.55 = .10d + .25q}$

equation

Using your equation or system of equations, algebraically determine the number of quarters Mr. Rogers deposited in the machine.

$17.55 = .10d + .25q$

$\boxed{\text{quarters: } 57}$

$1d + 1q = .35$

$.25 - .1 = .15$ → change

$59q = 14.75$

$31d = 3.1$

> 17.85

.3 over

$\frac{.3}{.15} = 2$ → 57q 33d

Mr. Rogers' neighbor replaced each one of his dimes with a quarter. If he uses all of his coins, determine if Mr. Rogers would have enough money to buy a cardigan priced at $20.98 if he must also pay an 8% sales tax. Justify your answer.

$20.98 + 8\% = 22.6584$

He would not have enough. With taxes the total would be $22.6584 which is more than he has.